

GUI Front End for RASOrbit - 2

6-19-2026

From the 9th Input Page on there can be four branches:

**SI (Metric) Units
Vertical Launch Mode**

**English Units
Vertical Launch Mode**

**SI (Metric) Units
Conventional Flight Mode**

**English Units
Conventional Flight Mode**

On the 10th Input Page One of Three Rocket Engine Model can be Selected:

- 0 - No Rocket Engine, Glider or Coasting Rocket**
- 1 - Rocket Engine Model using Chamber Pressure**
- 2 - Thrust Time History Rocket Engine Model**

English Units selected

Vertical Launch Mode

**Rocket Engine Model
Selections will be
Detailed on Slides**

9th Input Page – English Units

Vertical Launch ☒

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

Conventional Flight ☐

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

9th Input Page (Cont'd) – English Units – Vertical Launch Mode

Vertical Launch ☒

Conventional Flight ☐

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

Then all this appears

Launch Azimuth (deg)

(True North 0 deg,
East 90 deg,
South 180 deg)



Allowable Values:
0.0000 deg to 359.9999 deg.

Use 4 digits, .0000

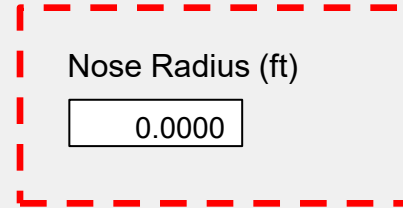
**Do not allow 360 deg,
360 deg should be entered as 0 deg.
Maybe have a pop-up warning
message.**

Do not allow negative numbers.

Nose Stagnation Heating Model

Yes ☐ No ☐

Nose Radius (ft)



If Nose Stagnation Heating Model = Yes Display This

If Nose Stagnation Heating Model = No Do Not Display This

Start with Box Blank.

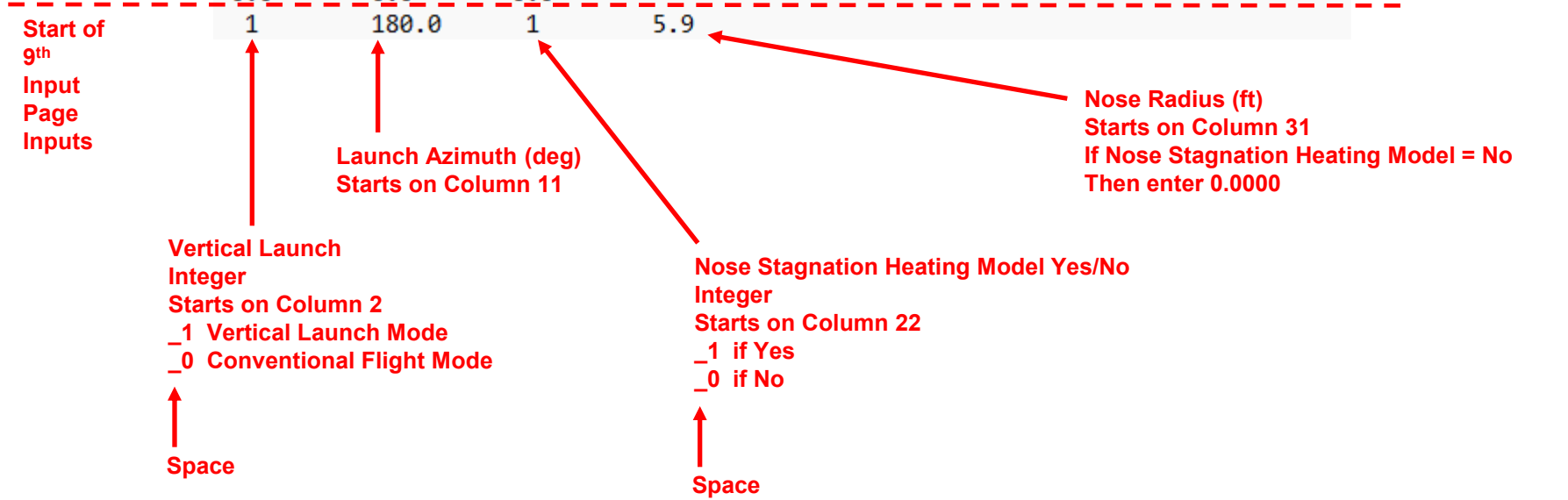
Allow User to dynamically use .0000 to as low as .0

Maximum value 9000000.0

Do not allow negative numbers

9th Input Page (Cont'd) – English Units – Vertical Launch Mode

2			
0.0	0.0		
0.0	0.0		
0.0	0.0		
0.0	0.0	0.0	
1	180.0	1	5.9



9th Input Page (Cont'd) – English Units – Vertical Launch Mode

Vertical Launch ☒

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

Conventional Flight ☐

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

Launch Azimuth (deg)

(True North 0 deg,
East 90 deg,
South 180 deg)

Nose Stagnation Heating Model

Yes ☐ No ☐

Nose Radius (ft)

-90.0000 deg to
90.0000 deg.
4 digits, .0000

Initial
Altitude (ft)
(Above Sea
Level)

Start with Box
Blank.
Allow User
to
dynamically
use .0000 to
.0
Limit is
9000000.0

Initial
Velocity
(ft/sec)

Start with Box
Blank. If Vertical
Launch Mode
Autofill This
with 0.0000
Allow User to
dynamically
use .0000 to
.0 Limit is
9000000.0
ft/sec. No Negative
Numbers.

Geodetic
Latitude
(deg)

-90.0000 deg
to 90.0000 deg.
4 digits, .0000
Do not allow
over 90.0000,
or lower than
-90.0000

Longitude
(deg)

0.0000 deg to
359.9999 deg.
4 digits, .0000
Do not allow 360
deg, 360 deg
should be entered
as 0 deg. Maybe
have a pop-up
warning message
for this.

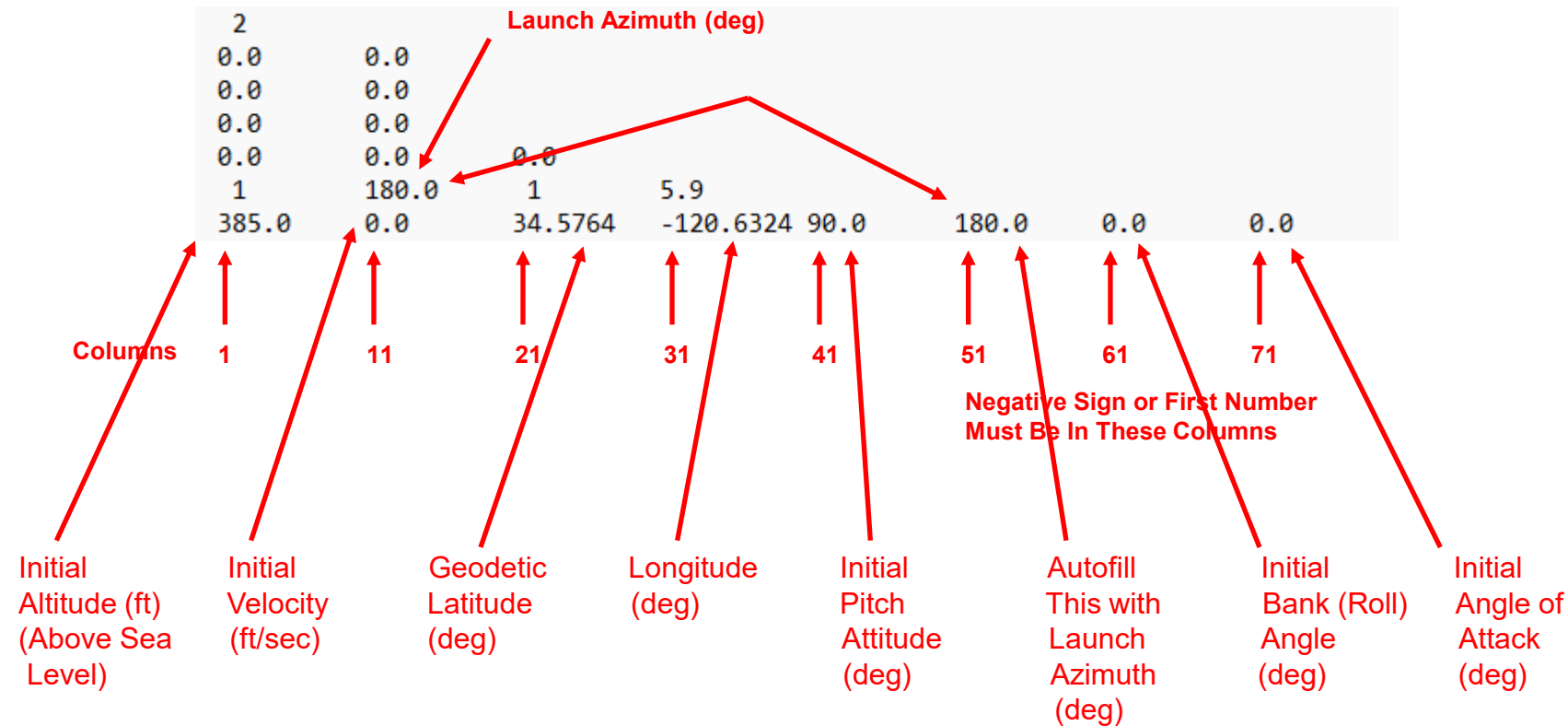
Initial
Pitch
Attitude
(deg)

Initial
Bank (Roll)
Angle
(deg)

If Vertical Launch Mode,
autofill:
Initial
Pitch Attitude = 90.0000
Bank Angle = 0.0000
Angle of Attack = 0.0000

Initial
Angle of
Attack
(deg)

9th Input Page (Cont'd) – English Units – Vertical Launch Mode



If Vertical Launch Mode,
autofill:
Initial Pitch Attitude = 90.0000 deg
Initial Bank (Roll) Angle = 0.0000 deg
Initial Angle of Attack = 0.0000 deg

9th Input Page Continued

Initial Altitude (ft) (Above Sea Level)	Initial Velocity (ft/sec)	Geodetic Latitude (deg)	Longitude (deg)	Initial Pitch Attitude (deg)	Initial Bank (Roll) Angle (deg)	Initial Angle of Attack (deg)
		0.0000	0.0000	0.0000	0.0000	0.0000

Integration Time Step (sec)	Total Time for Trajectory Run (sec)	Printout Output Rate (sec)
0.0000		0.0000

4 digits, .0000
Cannot be a
negative
number.
Cannot be
zero.

Start with Box
Blank.
Allow User
to
dynamically
use .0000 to
.0
Limit is
10000.000
Cannot be a
negative
number.

4 digits, .0000
Cannot be a
negative
number.
Cannot be
zero. Cannot
be smaller
than
Integration
Time Step.

Save

Cancel

9th Input Page – English Units – Vertical Launch Mode

2								
0.0	0.0							
0.0	0.0							
0.0	0.0							
0.0	0.0	0.0						
1	180.0	1	5.9					
385.0	0.0	34.5764	-120.6324	90.0	180.0	0.0	0.0	
0.01	578.0	1.0						

Columns

1

11

21

Integration
Time
Step
(sec)

Total
Time
for
Trajectory
Run
(sec)

Printout
Output
Rate
(sec)

English Units selected

**Conventional Flight Mode
Changes in Blue**

**Rocket Engine Model
Selections will be
Detailed on Slides**

9th Input Page (Cont'd) – English Units – Conventional Flight Mode

Vertical Launch ☐

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

Conventional Flight ☒

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

Then all this appears

Initial Heading (deg)

0.0000

(True North 0 deg,
East 90 deg,
South 180 deg)



Allowable Values:
0.0000 deg to 359.9999 deg.

Use 4 digits, .0000

**Do not allow 360 deg,
360 deg should be entered as 0 deg.
Maybe have a pop-up warning
message.**

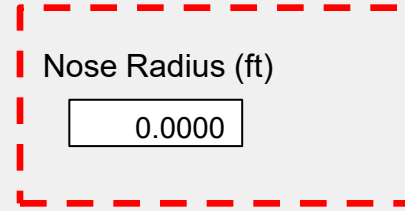
Do not allow negative numbers.

Nose Stagnation Heating Model

Yes ☐ No ☐

Nose Radius (ft)

0.0000



If Nose Stagnation Heating Model = Yes Display This

If Nose Stagnation Heating Model = No Do Not Display This

Start with Box Blank.

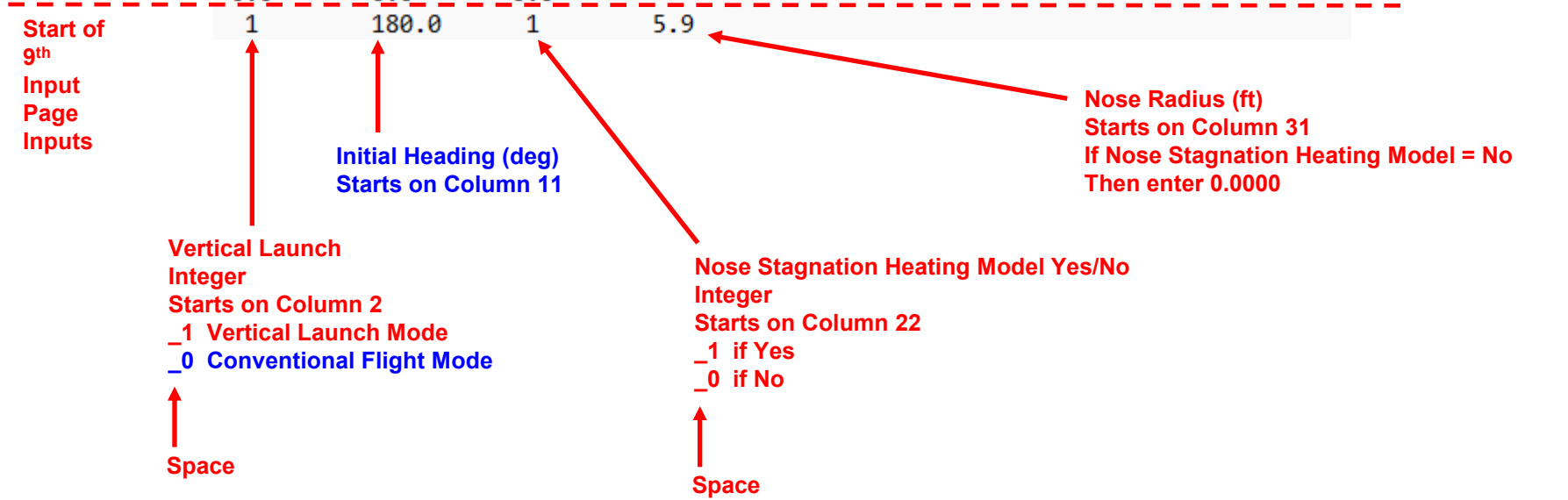
Allow User to dynamically use .0000 to as low as .0

Maximum value 9000000.0

Do not allow negative numbers

9th Input Page (Cont'd) – English Units – Conventional Flight Mode

2			
0.0	0.0		
0.0	0.0		
0.0	0.0		
0.0	0.0	0.0	
1	180.0	1	5.9



9th Input Page (Cont'd) – English Units – Conventional Flight Mode

Vertical Launch ☐

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

Conventional Flight ☒

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

Initial Heading (deg)

(True North 0 deg,
East 90 deg,
South 180 deg)

Nose Stagnation Heating Model

Yes ☐ No ☐

Nose Radius (ft)

Initial
Altitude (ft)
(Above Sea
Level)

Initial
Velocity
(ft/sec)

Geodetic
Latitude
(deg)

Longitude
(deg)

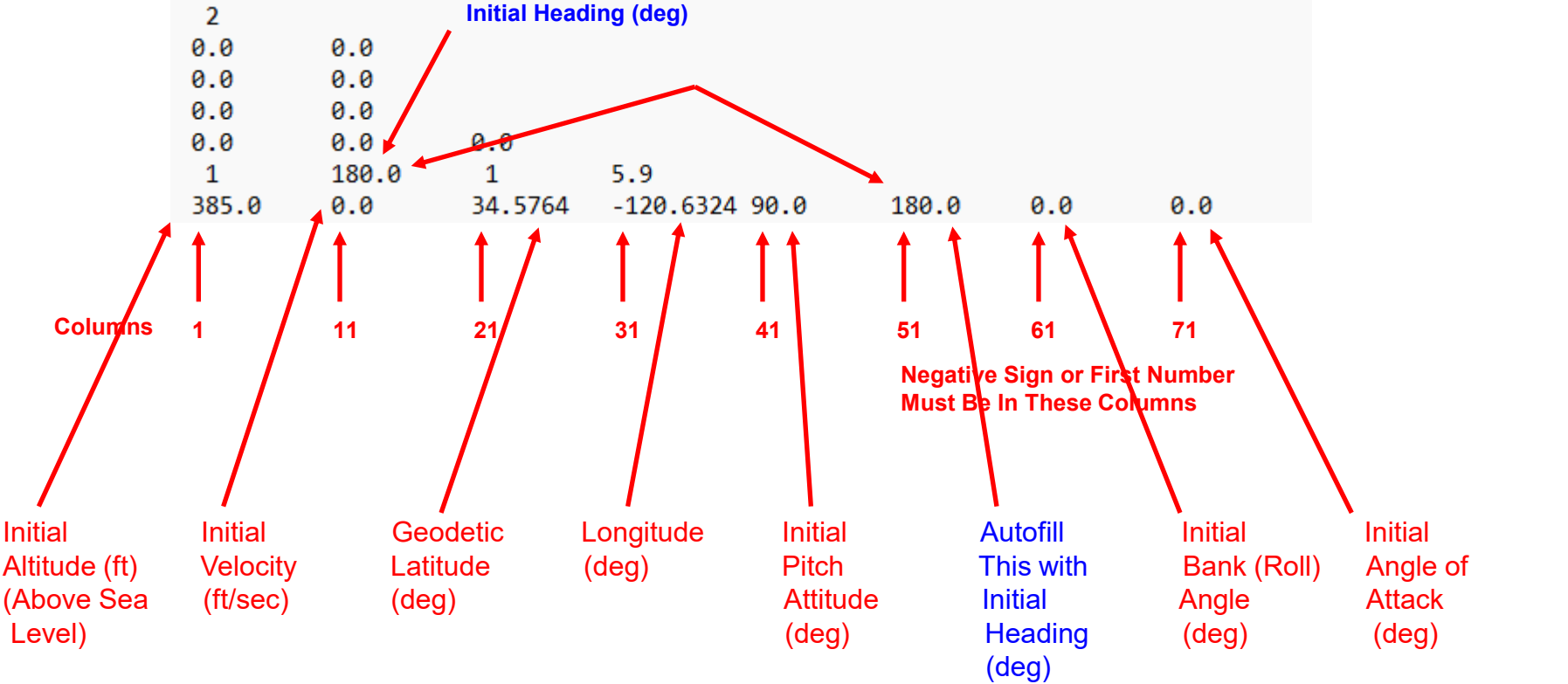
Initial
Pitch
Attitude
(deg)

Initial
Bank (Roll)
Angle
(deg)

Initial
Angle of
Attack
(deg)

Allow User to
dynamically
use .0000 to
.0 Limit is
9000000.0
ft/sec. No Negative
Numbers.

9th Input Page (Cont'd) – English Units – Conventional Flight Mode



English Units selected

**From This Point On It Doesn't
Matter Whether it is
Vertical Launch Mode
or
Conventional Flight Mode**

**Details Based On Which Rocket
Engine Model is Selected Will
Be Detailed on Slides.**

10th Input Page – English Units

For Stage 1: This will dynamically update. (Stage 1, Stage 2, etc.)

Stage Data

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
				0

Integer,
must be
0, 1, or 2

Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 Cannot be a negative number.	Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 Cannot be a negative number.	Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 Cannot be a negative number.	Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 Cannot be a negative number.
--	--	--	--

0 - No Rocket Engine, Glider or Coasting Rocket
1 - Rocket Engine Model using Chamber Pressure
2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

10th Input Page (Cont'd) – English Units

For Stage 1: ← This will dynamically update. (Stage 1, Stage 2, etc.)
← Make these Bold
Stage Data

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
				0

- 0 - No Rocket Engine, Glider or Coasting Rocket
- 1 - Rocket Engine Model using Chamber Pressure
- 2 - Thrust Time History Rocket Engine Model

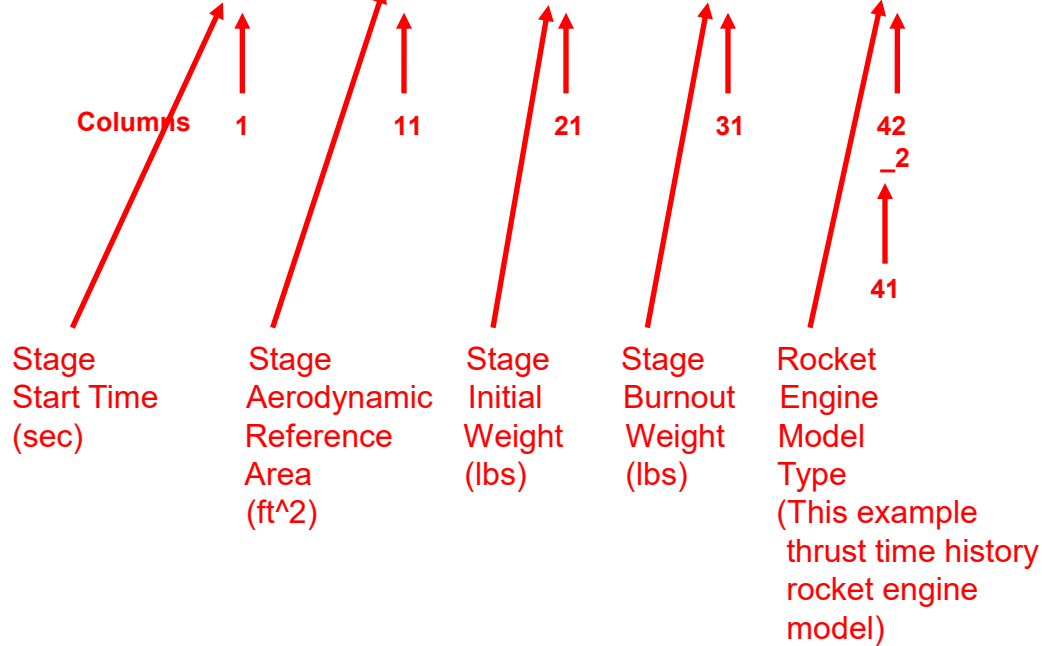
Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

If Rocket Engine Model Type = 0
(No Rocket Engine, Glider
or Coasting Rocket),

Then There Are No Further Inputs
Here.

10th Input Page (Cont'd) – English Units

0.0	23.3997	79850.0	34020.0	2
61.38	12.2431	14.69	0	
61.39	21.8264	28965.0	15212.0	2
64.4	5.0688	14.69	0	
125.8	14.748	13409.0	4776.0	2
74.5	6.268	0.0	0	
510.0	14.748	3004.0	1305.0	2
67.3	2.3371	0.0	0	
0.0	23.3997	79850.0	34020.0	2



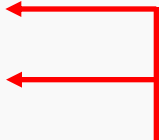
10th Input Page (Cont'd) – English Units

If Rocket Engine Model Type = 0
(No Rocket Engine, Glider
or Coasting Rocket),

Then These Lines are Empty.



0.0	23.3997	79850.0	34020.0	0
61.38	12.2431	14.69	0	
61.39	21.8264	28965.0	15212.0	0
64.4	5.0688	14.69	0	
125.8	14.748	13409.0	4776.0	0
74.5	6.268	0.0	0	
510.0	14.748	3004.0	1305.0	0
67.3	2.3371	0.0	0	



These are the

Stage	Stage	Stage	Stage	Rocket
Start Time	Aerodynamic	Initial	Burnout	Engine
(sec)	Reference	Weight	Weight	Model
	Area	(lbs)	(lbs)	Type
	(ft^2)			

Inputs

10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 1

For Stage 1: This will dynamically update. (Stage 1, Stage 2, etc.)

Stage Data Make these Bold

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
				0

If Rocket Engine Model Type = 1
(Rocket Engine Model using
Chamber Pressure),
Then These Inputs.

- 0 - No Rocket Engine, Glider or Coasting Rocket
- 1 - Rocket Engine Model using Chamber Pressure
- 2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Throat Area (ft^2)	Nozzle Expansion Ratio (n.d.)	Nozzle Divergence Half Angle (deg)	Burn Time (sec)	Reference Thrust (lbs)	Reference Specific Impulse (sec)	Reference Chamber Pressure (lbs/in^2)	Reference Atmospheric Pressure (lbs/in^2)
	0.0000	0.0000					0.0000

Ratio of Specific Heats (n.d.)	Thrust Coefficient Ratio (n.d.)	Nozzle Type 1 – Conical 2 – Bell	Negative Thrust 0 – Positive Thrust 1 – Negative Thrust
0.0000	0.0000	1	0

For Blank Boxes, Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0

For Boxes with 0.0000, 4 digits, .0000

Have display start with Zero (Positive Thrust). User has to change it (if they want to).

Integer

For Stage 1:

10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 1 Rocket Engine Model using Chamber Pressure

Stage Data

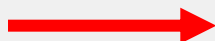
Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft ²)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
				0

0 - No Rocket Engine, Glider or Coasting Rocket
1 - Rocket Engine Model using Chamber Pressure
2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Throat Area (ft ²)	Nozzle Expansion Ratio (n.d.)	Nozzle Divergence Half Angle (deg)	Burn Time (sec)	Reference Thrust (lbs)	Reference Specific Impulse (sec)	Reference Chamber Pressure (lbs/in ²)	Reference Atmospheric Pressure (lbs/in ²)
	0.0000	0.0000					0.0000

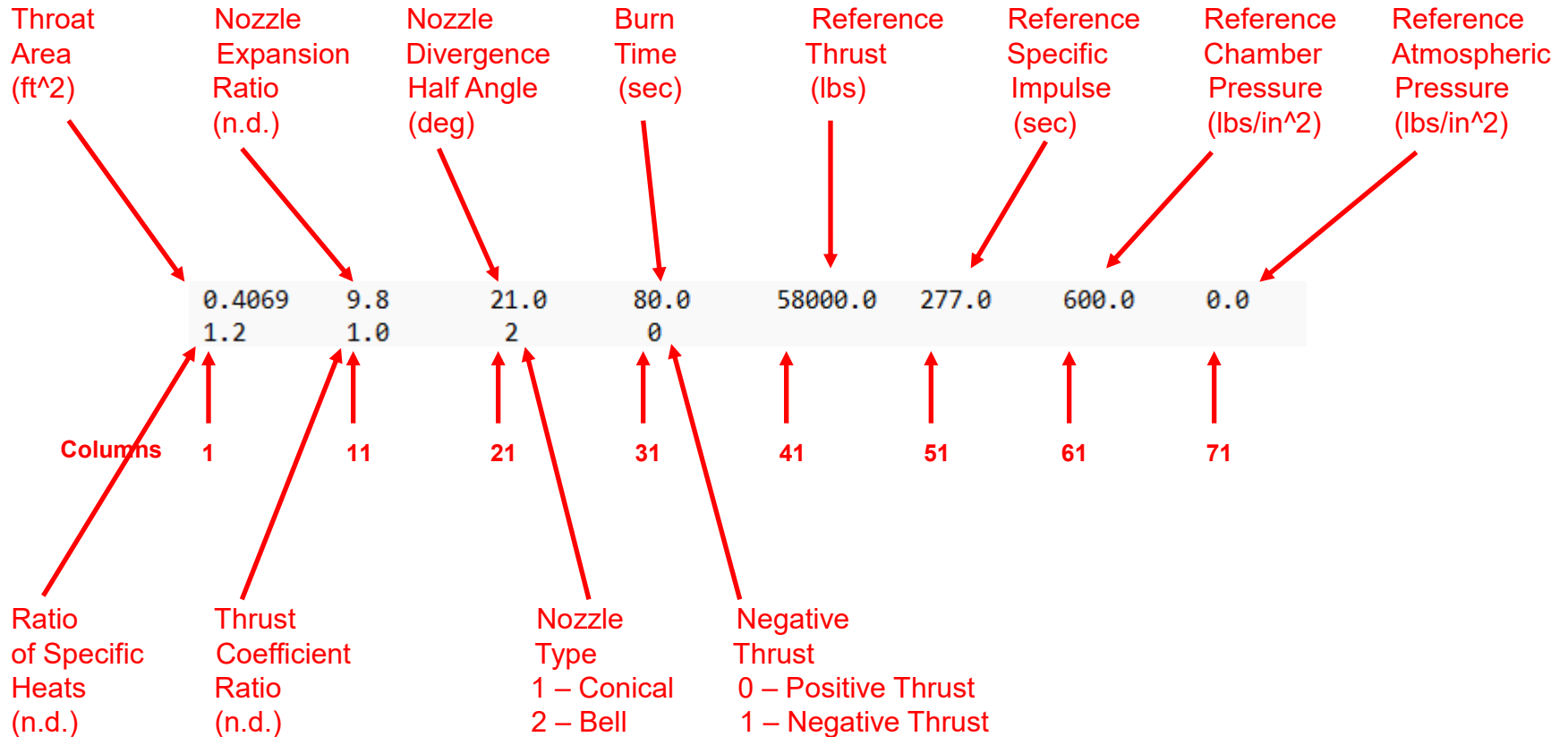
Ratio of Specific Heats (n.d.)	Thrust Coefficient Ratio (n.d.)	Nozzle Type 1 – Conical 2 – Bell	Negative Thrust 0 – Positive Thrust 1 – Negative Thrust
0.0000	0.0000	1	0



Save

Cancel

10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 1 Rocket Engine Model using Chamber Pressure



This was Rocket Engine Model = 1 Rocket Engine Model using Chamber Pressure

If Rocket Engine Model Type = 0 No Rocket Engine, Glider or Coasting Rocket),

Then These Input Lines Would Not Be Present.

For Stage 1:

10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 1 Rocket Engine Model using Chamber Pressure

Stage Data

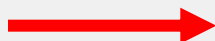
Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft ²)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
				0

0 - No Rocket Engine, Glider or Coasting Rocket
1 - Rocket Engine Model using Chamber Pressure
2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Throat Area (ft ²)	Nozzle Expansion Ratio (n.d.)	Nozzle Divergence Half Angle (deg)	Burn Time (sec)	Reference Thrust (lbs)	Reference Specific Impulse (sec)	Reference Chamber Pressure (lbs/in ²)	Reference Atmospheric Pressure (lbs/in ²)
	0.0000	0.0000					0.0000

Ratio of Specific Heats (n.d.)	Thrust Coefficient Ratio (n.d.)	Nozzle Type 1 – Conical 2 – Bell	Negative Thrust 0 – Positive Thrust 1 – Negative Thrust
0.0000	0.0000	1	0



Save

Cancel

10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 2

For Stage 1: This will dynamically update. (Stage 1, Stage 2, etc.)

Stage Data Make these Bold

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
				0

If Rocket Engine Model Type = 2
(Thrust Time History Rocket Engine
Model),

- 0 - No Rocket Engine, Glider or Coasting Rocket
- 1 - Rocket Engine Model using Chamber Pressure
- 2 - Thrust Time History Rocket Engine Model

Then These Inputs.

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Burn Time (sec)	Nozzle Exit Area (ft^2)	Reference Atmospheric Pressure (lbs/ft^2)	Negative Thrust 0 – Positive Thrust 1 – Negative Thrust
		0.0000	0

For Blank Boxes, Start with Box Blank. Allow User
to dynamically use .0000 to .0 Limit is 9000000.0

For Boxes with 0.0000, 4 digits, .0000

Integer

Have display start with Zero (Positive Thrust).
User has to change it (if they want to).

Save

Cancel

10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 2
Thrust Time History Rocket Engine Model

These are the

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
------------------------------	---	-------------------------------------	-------------------------------------	-----------------------------------

2		Inputs					
0.0	0.0						
0.0	0.0						
0.0	0.0						
0.0	0.0	0.0					
1	180.0	1	5.9				
385.0	0.0	34.5764	-120.6324	90.0	180.0	0.0	0.0
0.01	578.0	1.0					
0.0	23.3997	79850.0	34020.0	2			
61.38	12.2431	14.69	0				

Columns

1

11

21

31

32

0

Burn Time (sec)

Nozzle Exit Area (ft^2)

Reference Atmospheric Pressure (lbs/ft^2)

Negative Thrust
0 – Positive Thrust
1 – Negative Thrust

10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 2
Thrust Time History Rocket Engine Model

Repeated for 4 Stages

0.0	23.3997	79850.0	34020.0	2	Stage – 1
61.38	12.2431	14.69	0		
61.39	21.8264	28965.0	15212.0	2	Stage – 2
64.4	5.0688	14.69	0		
125.8	14.748	13409.0	4776.0	2	Stage – 3
74.5	6.268	0.0	0		
510.0	14.748	3004.0	1305.0	2	Stage - 4
67.3	2.3371	0.0	0		

For Stage 1: **10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 0**
No Rocket Engine, Glider or Coasting Rocket

Stage Data

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
				0

**If Rocket Engine Model Type = 0
(No Rocket Engine, Glider
or Coasting Rocket),**

Then These Inputs are Not Displayed.

0 - No Rocket Engine, Glider or Coasting Rocket
1 - Rocket Engine Model using Chamber Pressure
2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Throat	Nozzle	Nozzle	Burn	Reference	Reference	Reference	Reference
Area	Expansion	Divergence	Time	Thrust	Specific	Chamber	Atmospheric
(ft^2)	Ratio	Half Angle	(sec)	(lbs)	Impulse	Pressure	Pressure
(n.d.)	(n.d.)	(deg)			(sec)	(lbs/in^2)	(lbs/in^2)
 	 0.0000 	 0.0000 	 	 	 	 	 0.0000

Ratio	Thrust	Nozzle	Negative
of Specific	Coefficient	Type	Thrust
Heats	Ratio	1 Conical	0 Positive Thrust
(n.d.)	(n.d.)	2 Bell	1 Negative Thrust
 0.0000 	 0.0000 	 1 	 0

Save

Cancel

10th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 0

No Rocket Engine, Glider or Coasting Rocket

For Stage 1:

Stage Data

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
				0

If Rocket Engine Model Type = 0
(No Rocket Engine, Glider
or Coasting Rocket),

Then These Inputs are Not Displayed.

- 0 - No Rocket Engine, Glider or Coasting Rocket
1 - Rocket Engine Model using Chamber Pressure
2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Burn	Nozzle	Reference	Negative
Time	Exit	Atmospheric	Thrust
(sec)	Area	Pressure	0 – Positive Thrust
	(ft^2)	(lbs/ft^2)	1 – Negative Thrust
 	 	 0.0000 	 0

Save

Cancel

11th Input Page – English Units – Rocket Engine Model Type = 0
No Rocket Engine, Glider or Coasting Rocket

Chamber Pressure Time History

Number of Chamber Pressure Time History Points

Time (sec)

Chamber Pressure (lbs/in^2)

If Rocket Engine Model Type = 0
(No Rocket Engine, Glider or
Coasting Rocket)

11th Input Page is Not Displayed.
Jump to 12th Input Page.

Save

Cancel

11th Input Page – English Units – Rocket Engine Model Type = 1
Chamber Pressure Time History

Chamber Pressure Time History

Number of Chamber Pressure Time History Points

Minimum of 2, Maximum of 40, is an Integer

Time (sec)

Chamber Pressure (lbs/in^2)

Will dynamically update based on number of Chamber Pressure Time History Points. 8 Boxes per Row until Final Row, Final Row will have Number to Match Number of Inputs.

Blank Boxes, Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 No Negative Numbers.

Save

Cancel

11th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 1
Chamber Pressure Time History

First Time Must Be Zero

Number of Chamber Pressure Time History Points
Integer
If > 9, then two digits (38)
If < 10, then _3 _8 _9 etc., put one space in front of Integer

Time (sec)

Chamber Pressure (lbs/in^2)

3		
0.0	79.9	80.0
600.0	600.0	0.0

Columns 1 11 21

11th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 2
Thrust Time History (Thrust Curve)

Thrust Time History

Number of Thrust Time History Points

38

Minimum of 2, Maximum of 40, is an Integer

Time (sec)

Thrust (lbs)

First Time Must Be Zero

Will dynamically update based on number of Chamber Pressure Time History Points. 8 Boxes per Row until Final Row, Final Row will have Number to Match Number of Inputs.

Blank Boxes, Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 No Negative Numbers.

Save

Cancel

11th Input Page (Cont'd) – English Units – Rocket Engine Model Type = 2
Thrust Time History (Thrust Curve)

First Time Must Be Zero

Number of Thrust Time History Points
Integer
If > 9, then two digits (38)
If < 10, then _8 _9 etc., put one space in front of Integer

	38							
Time (sec)	0.0	0.71	5.40	15.94	31.76	47.59	52.86	57.28
	61.38	61.39	66.99	72.07	77.14	87.30	97.46	112.69
	124.40	125.79	125.8	140.8	155.8	160.8	165.8	180.8
	191.59	195.8	200.3	510.0	510.01	513.1	525.0	533.0
	540.0	567.0	568.6	574.8	577.3	578.0		
Thrust (lbs)	102186.7	199786.7	198788.4	203779.8	200170.6	201860.0	200093.9	45822.8
	0.0	42654.1	51364.6	55211.3	56928.0	56355.7	50379.1	44211.8
	30446.6	0.0	22250.0	37105.0	42949.0	43846.0	42949.0	37632.0
	28590.0	2308.0	0.0	0.0	5213.0	6133.3	7783.0	7783.0
	8255.5	8255.5	6000.0	4000.0	0.0	0.0		
Columns	1	11	21	31	41	51	61	71

Example
Rocket Engine Model Type = 0
(No Rocket Engine, Glider or Coasting Rocket)

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type
0.0	2690.0	208790.0	208790.0	0

Rocket Engine Model Type = 0
(No Rocket Engine, Glider or Coasting Rocket)

No Rocket Engine Inputs.
No Rocket Engine Chamber Pressure or Thrust Time History Inputs.

Angle of Attack or Pitch
Attitude Time History
Inputs (will be detailed
on later slides).

20	1						
0.0	330.000	340.000	450.000	500.000	600.000	880.000	894.000
900.000	901.000	910.000	911.0	940.000	941.000	980.000	990.000
1080.000	1081.000	1120.000	1075.000				
40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0
43.0	43.0	42.0	42.0	40.0	40.0	40.0	40.0
39.0	39.0	39.0	39.0				
0.0	0.0	80.0	80.0	75.0	70.0	65.0	-85.0
-85.0	-85.0	-85.0	-85.0	-60.0	-60.0	-70.0	-55.0
-65.0	-65.0	-65.0	-65.0				

End of Input File

(Rocket Engine Model using Chamber Pressure)

Chamber Pressure Time History Inputs

Angle of Attack or Pitch
Attitude Time History
Inputs (will be detailed
on later slides).

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type			
0.0	200.0	33300.0	15000.0	1			
0.4069	9.8	21.0	80.0	58000.0	277.0	600.0	0.0
1.2	1.0	2	0				
3							
0.0	79.9	80.0					
600.0	600.0	0.0					
10	1						
0.0	5.0	6.5	20.0	30.0	190.0	200.0	280.0
290.0	1000.0						
0.0	0.0	15.0	15.0	5.0	10.0	18.0	18.0
10.0	10.0						
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0						

End of Input File

4 Stages

Angle of Attack or Pitch
Attitude Time History
Inputs (will be detailed
on later slides).

Stage Start Time (sec)	Stage Aerodynamic Reference Area (ft^2)	Stage Initial Weight (lbs)	Stage Burnout Weight (lbs)	Rocket Engine Model Type			
0.0	23.3997	79850.0	34020.0	2			
61.38	12.2431	14.69	0				
61.39	21.8264	28965.0	15212.0	2			
64.4	5.0688	14.69	0				
125.8	14.748	13409.0	4776.0	2			
74.5	6.268	0.0	0				
510.0	14.748	3004.0	1305.0	2			
67.3	2.3371	0.0	0				
38							
0.0	0.71	5.40	15.94	31.76	47.59	52.86	57.28
61.38	61.39	66.99	72.07	77.14	87.30	97.46	112.69
124.40	125.79	125.8	140.8	155.8	160.8	165.8	180.8
191.59	195.8	200.3	510.0	510.01	513.1	525.0	533.0
540.0	567.0	568.6	574.8	577.3	578.0		
102186.7	199786.7	198788.4	203779.8	200170.6	201860.0	200093.9	45822.8
0.0	42654.1	51364.6	55211.3	56928.0	56355.7	50379.1	44211.8
30446.6	0.0	22250.0	37105.0	42949.0	43846.0	42949.0	37632.0
28590.0	2308.0	0.0	0.0	5213.0	6133.3	7783.0	7783.0
8255.5	8255.5	6000.0	4000.0	0.0	0.0		
17	1						
0.0	2.0	3.0	4.0	8.0	10.0	11.0	12.0
13.0	14.0	15.0	23.0	30.0	57.0	144.0	192.0
578.0							
0.0	0.0	-3.3	-7.0	-7.0	-6.0	-4.0	0.0
0.6	0.7	0.7	0.5	0.4	0.191	0.1	0.06
0.06							
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	End of Input File						

Trajectory Control

Angle of Attack and Bank Angle ☒ Pitch Attitude and Bank Angle ☐

12th Input Page (Cont'd) – English Units – Angle of Attack and Bank Angle Time History

Trajectory Control

Angle of Attack and Bank Angle ☒ Pitch Attitude and Bank Angle ☐

Then this appears

Number of Angle of Attack and Bank Angle Time History Points Minimum of 2, Maximum of 40, is an Integer

12th Input Page (Cont'd) – English Units – Angle of Attack and Bank Angle Time History

Trajectory Control

Angle of Attack and Bank Angle ☒ Pitch Attitude and Bank Angle ☐

Number of Angle of Attack and Bank Angle Time History Points Minimum of 2, Maximum of 40, is an Integer

Then all this appears

Time (sec)

First Time Must Be Zero

Angle of Attack (deg)

0.0000	0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000	0.0000

Bank Angle (Roll Angle) (deg)

0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Will dynamically update based on number of Time History Points. 8 Boxes per Row until Final Row, Final Row will have Number to Match Number of Inputs.

Blank Boxes, Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 No Negative Numbers.

For Boxes with 0.0000, 4 digits, .0000

Angle of Attack and Bank Angle Limits are -90.0000 and 90.0000

Save

Cancel

12th Input Page (Cont'd) – English Units – Angle of Attack and Bank Angle Time History

Number of Angle of Attack and Bank Angle Time History Points
Integer
If > 9, then two digits (38)
If < 10, then _8 _9 etc., put one space in front of Integer

For Angle of Attack and Bank Angle Control
This is _1 (An Integer)

First Time Must Be Zero

Time (sec)

Angle of Attack (deg)

Bank Angle (Roll Angle) (deg)

Columns

17	1						
0.0	2.0	3.0	4.0	8.0	10.0	11.0	12.0
13.0	14.0	15.0	23.0	30.0	57.0	144.0	192.0
578.0							
0.0	0.0	-3.3	-7.0	-7.0	-6.0	-4.0	0.0
0.6	0.7	0.7	0.5	0.4	0.191	0.1	0.06
0.06							
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0							
↑	↑	↑	↑	↑	↑	↑	↑
1	11	21	31	41	51	61	71

12th Input Page (Cont'd) – English Units – Pitch Attitude and Bank Angle Time History

Trajectory Control

Angle of Attack and Bank Angle ☐

Pitch Attitude and Bank Angle ☒

12th Input Page (Cont'd) – English Units – Pitch Attitude and Bank Angle Time History

Trajectory Control

Angle of Attack and Bank Angle ☐ Pitch Attitude and Bank Angle ☒

Then this appears

Number of Pitch Attitude and Bank Angle Time History Points Minimum of 2, Maximum of 40, is an Integer

12th Input Page (Cont'd) – English Units – Pitch Attitude and Bank Angle Time History

Trajectory Control

Angle of Attack and Bank Angle ☐ Pitch Attitude and Bank Angle ☒

Number of Pitch Attitude and Bank Angle Time History Points Minimum of 2, Maximum of 40, is an Integer

Then all this appears

Time (sec)

First Time Must Be Zero

Pitch Attitude (deg)

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

Bank Angle (Roll Angle) (deg)

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

0.0000

Will dynamically update based on number of Time History Points. 8 Boxes per Row until Final Row, Final Row will have Number to Match Number of Inputs.

Blank Boxes, Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 No Negative Numbers.

For Boxes with 0.0000, 4 digits, .0000

Pitch Attitude and Bank Angle Limits are -90.0000 and 90.0000

Save

Cancel

12th Input Page (Cont'd) – English Units – Pitch Attitude and Bank Angle Time History

Number of Pitch Attitude and Bank Angle Time History Points
Integer
If > 9, then two digits (38)
If < 10, then _8 _9 etc., put one space in front of Integer

For Pitch Attitude and Bank Angle Control
This is _0 (An Integer)

First Time Must Be Zero

Time (sec)

Pitch Attitude (deg)

Bank Angle (Roll Angle) (deg)

Columns

17	1						
0.0	2.0	3.0	4.0	8.0	10.0	11.0	12.0
13.0	14.0	15.0	23.0	30.0	57.0	144.0	192.0
578.0							
0.0	0.0	-3.3	-7.0	-7.0	-6.0	-4.0	0.0
0.6	0.7	0.7	0.5	0.4	0.191	0.1	0.06
0.06							
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0							
↑	↑	↑	↑	↑	↑	↑	↑
1	11	21	31	41	51	61	71

**SI (Metric) Units selected
Changes in Blue**

Vertical Launch Mode

**Rocket Engine Model
Selections will be
Detailed on Slides**

9th Input Page (Cont'd) – SI (Metric) Units – Vertical Launch Mode

Vertical Launch ☒

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

Conventional Flight ☐

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

Then all this appears

Launch Azimuth (deg)

(True North 0 deg,
East 90 deg,
South 180 deg)



Allowable Values:
0.0000 deg to 359.9999 deg.

Use 4 digits, .0000

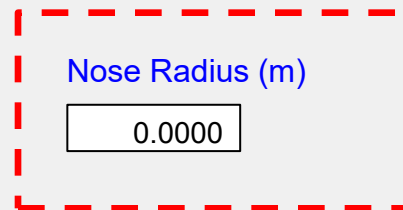
**Do not allow 360 deg,
360 deg should be entered as 0 deg.
Maybe have a pop-up warning
message.**

Do not allow negative numbers.

Nose Stagnation Heating Model

Yes ☐ No ☐

Nose Radius (m)



If Nose Stagnation Heating Model = Yes Display This

If Nose Stagnation Heating Model = No Do Not Display This

Start with Box Blank.

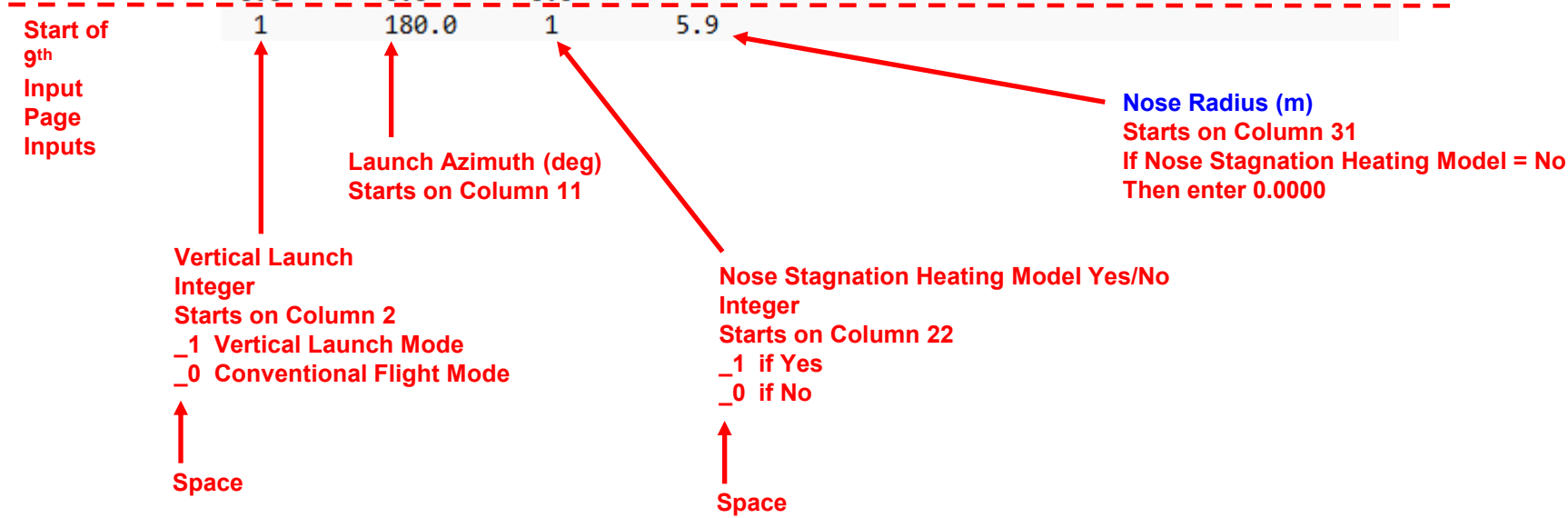
Allow User to dynamically use .0000 to as low as .0

Maximum value 9000000.0

Do not allow negative numbers

9th Input Page (Cont'd) – SI (Metric) Units – Vertical Launch Mode

2			
0.0	0.0		
0.0	0.0		
0.0	0.0		
0.0	0.0	0.0	
1	180.0	1	5.9



9th Input Page (Cont'd) – SI (Metric) Units – Vertical Launch Mode

Vertical Launch ☒

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

Conventional Flight ☐

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

Launch Azimuth (deg)

(True North 0 deg,
East 90 deg,
South 180 deg)

Nose Stagnation Heating Model

Yes ☐ No ☐

Nose Radius (m)

-90.0000 deg to
90.0000 deg.
4 digits, .0000

Initial
Altitude (m)
(Above Sea
Level)

Start with Box
Blank.
Allow User
to
dynamically
use .0000 to
.0
Limit is
9000000.0

Initial
Velocity
(m/sec)

Start with Box
Blank. If Vertical
Launch Mode
Autofill This
with 0.0000
Allow User to
dynamically
use .0000 to
.0 Limit is
9000000.0
ft/sec. No Negative
Numbers.

Geodetic
Latitude
(deg)

-90.0000 deg
to 90.0000 deg.
4 digits, .0000
Do not allow
over 90.0000,
or lower than
-90.0000

Longitude
(deg)

0.0000 deg to
359.9999 deg.
4 digits, .0000
Do not allow 360
deg, 360 deg
should be entered
as 0 deg. Maybe
have a pop-up
warning message
for this.

Initial
Pitch
Attitude
(deg)

Initial
Bank (Roll)
Angle
(deg)

If Vertical Launch Mode,
autofill:
Initial
Pitch Attitude = 90.0000
Bank Angle = 0.0000
Angle of Attack = 0.0000

Initial
Angle of
Attack
(deg)

9th Input Page (Cont'd) – SI (Metric) Units – Vertical Launch Mode

Launch Azimuth (deg)

0.0 0.0 0.0 0.0 0.0 1 385.0

0.0 180.0 1 34.5764 -120.6324 90.0 180.0 0.0 0.0

Columns

1

11

21

31

41

51

61

71

Initial
Altitude (m)
(Above Sea
Level)

Initial
Velocity
(m/sec)

Geodetic
Latitude
(deg)Longitude
(deg)

Initial
Pitch
Attitude
(deg)

Autofill
This with
Launch
Azimuth
(deg)

Initial
Bank (Roll)
Angle
(deg)

Initial
Angle of
Attack
(deg)

If Vertical Launch Mode, autofill:

Initial Pitch Attitude = 90.0000 deg

Initial Bank (Roll) Angle = 0.0000 deg

Initial Angle of Attack = 0.0000 deg

9th Input Page Continued

Initial Altitude (m) (Above Sea Level)	Initial Velocity (m/sec)	Geodetic Latitude (deg)	Longitude (deg)	Initial Pitch Attitude (deg)	Initial Bank (Roll) Angle (deg)	Initial Angle of Attack (deg)
		0.0000	0.0000	0.0000	0.0000	0.0000

Integration Time Step (sec)	Total Time for Trajectory Run (sec)	Printout Output Rate (sec)
0.0000		0.0000

4 digits, .0000
Cannot be a
negative
number.
Cannot be
zero.

Start with Box
Blank.
Allow User
to
dynamically
use .0000 to
.0
Limit is
10000.000
Cannot be a
negative
number.

4 digits, .0000
Cannot be a
negative
number.
Cannot be
zero. Cannot
be smaller
than
Integration
Time Step.

Save

Cancel

9th Input Page – SI (Metric) Units – Vertical Launch Mode

2								
0.0	0.0							
0.0	0.0							
0.0	0.0							
0.0	0.0	0.0						
1	180.0	1	5.9					
385.0	0.0	34.5764	-120.6324	90.0	180.0	0.0	0.0	
0.01	578.0	1.0						

Columns

1

11

21

Integration
Time
Step
(sec)

Total
Time
for
Trajectory
Run
(sec)

Printout
Output
Rate
(sec)

**SI (Metric) Units selected
Changes in Blue**

**Conventional Flight Mode
Changes in Blue**

**Rocket Engine Model
Selections will be
Detailed on Slides**

9th Input Page (Cont'd) – SI (Metric) Units – Conventional Flight Mode

Vertical Launch ☐

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

Conventional Flight ☒

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

Then all this appears

Initial Heading (deg)

0.0000

(True North 0 deg,
East 90 deg,
South 180 deg)



Allowable Values:
0.0000 deg to 359.9999 deg.

Use 4 digits, .0000

**Do not allow 360 deg,
360 deg should be entered as 0 deg.
Maybe have a pop-up warning
message.**

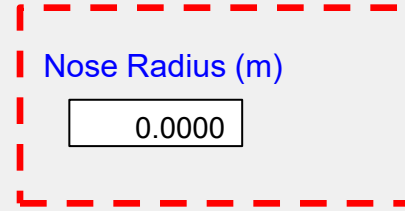
Do not allow negative numbers.

Nose Stagnation Heating Model

Yes ☐ No ☐

Nose Radius (m)

0.0000



If Nose Stagnation Heating Model = Yes Display This

If Nose Stagnation Heating Model = No Do Not Display This

Start with Box Blank.

Allow User to dynamically use .0000 to as low as .0

Maximum value 9000000.0

Do not allow negative numbers

9th Input Page (Cont'd) – SI (Metric) Units – Conventional Flight Mode

2			
0.0	0.0		
0.0	0.0		
0.0	0.0		
0.0	0.0	0.0	
1	180.0	1	5.9

Start of
9th
Input
Page
Inputs

Vertical Launch
Integer
Starts on Column 2
_1 Vertical Launch Mode
_0 Conventional Flight Mode

Space

Initial Heading (deg)
Starts on Column 11

Thrust Vectoring Gimbal Location of Last Stage

Percent of Thrust for Pitch Thrust Vectoring of Last Stage

Maximum Thrust Vector Angle of Last Stage

Nose Radius (m)
Starts on Column 31
If Nose Stagnation Heating Model = No
Then enter 0.0000

Nose Stagnation Heating Model Yes/No
Integer
Starts on Column 22
_1 if Yes
_0 if No

Space

9th Input Page (Cont'd) – SI (Metric) Units – Conventional Flight Mode

Vertical Launch ☐

(Ground Launch from Launch Site,
90 deg Pitch Attitude, Zero Initial Velocity,
Vertical Lift-Off)

Conventional Flight ☒

(Any Flight Condition, Including Air-Launch,
and Starting a Stage at Any Altitude and
Velocity)

Initial Heading (deg)

(True North 0 deg,
East 90 deg,
South 180 deg)

Nose Stagnation Heating Model

Yes ☐ No ☐

Nose Radius (m)

Initial
Altitude (m)
(Above Sea
Level)

Initial
Velocity
(m/sec)

Geodetic
Latitude
(deg)

Longitude
(deg)

Initial
Pitch
Attitude
(deg)

Initial
Bank (Roll)
Angle
(deg)

Initial
Angle of
Attack
(deg)

Allow User to
dynamically
use .0000 to
.0 Limit is
9000000.0
ft/sec. No Negative
Numbers.

9th Input Page (Cont'd) – SI (Metric) Units – Conventional Flight Mode

[illegible]

Columns

1

11

21

31

41

51

61

71

Initial
Altitude (m)
(Above Sea
Level)

Initial
Velocity
(m/sec)

Geodetic
Latitude
(deg)Longitude
(deg)

Initial
Pitch
Attitude
(deg)

Autofill
This with
Initial
Heading
(deg)

Initial
Bank (Roll)
Angle
(deg)

Initial
Angle of
Attack
(deg)

**Negative Sign or First Number
Must Be In These Columns**

**SI (Metric) Units selected
Changes in Blue**

**From This Point On It Doesn't
Matter Whether it is
Vertical Launch Mode
or
Conventional Flight Mode**

**Details Based On Which Rocket
Engine Model is Selected Will
Be Detailed on Slides.**

10th Input Page – SI (Metric) Units

For Stage 1: This will dynamically update. (Stage 1, Stage 2, etc.)

Stage Data

Stage Start Time (sec)	Stage Aerodynamic Reference Area (m^2)	Stage Initial Weight (kg)	Stage Burnout Weight (kg)	Rocket Engine Model Type
				0

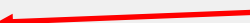
Integer,
must be
0, 1, or 2

Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 Cannot be a negative number.	Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 Cannot be a negative number.	Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 Cannot be a negative number.	Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 Cannot be a negative number.
--	--	--	--

0 - No Rocket Engine, Glider or Coasting Rocket
1 - Rocket Engine Model using Chamber Pressure
2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

10th Input Page (Cont'd) – SI (Metric) Units

For Stage 1:  This will dynamically update. (Stage 1, Stage 2, etc.)

 Make these Bold
Stage Data

Stage Start Time (sec)	Stage Aerodynamic Reference Area (m^2)	Stage Initial Weight (kg)	Stage Burnout Weight (kg)	Rocket Engine Model Type
				0

- 0 - No Rocket Engine, Glider or Coasting Rocket
- 1 - Rocket Engine Model using Chamber Pressure
- 2 - Thrust Time History Rocket Engine Model

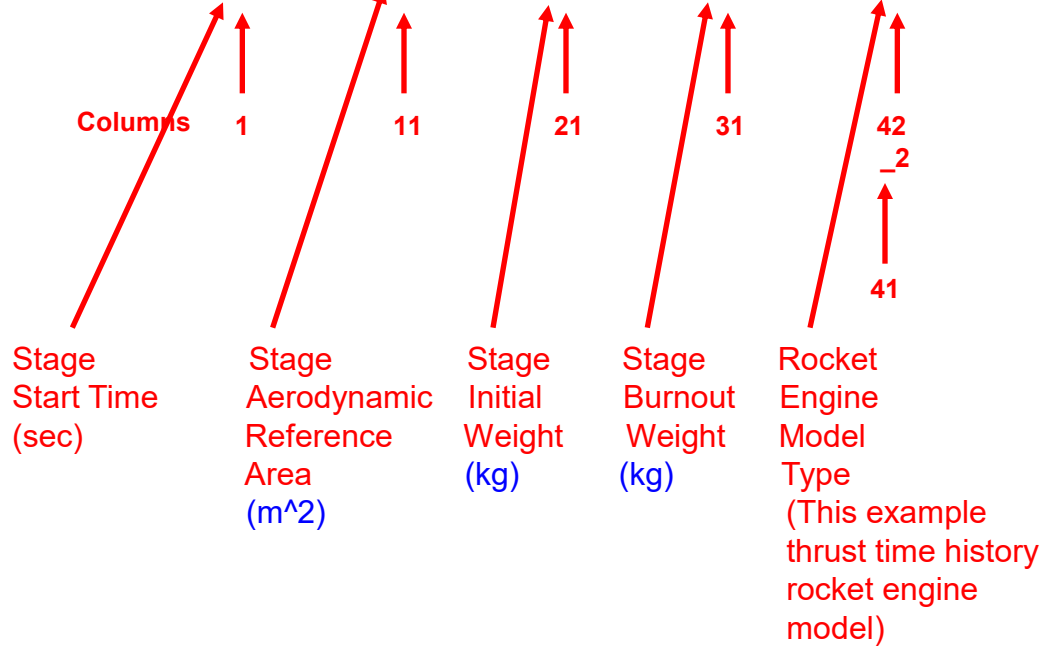
Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

If Rocket Engine Model Type = 0
(No Rocket Engine, Glider
or Coasting Rocket),

Then There Are No Further Inputs
Here.

10th Input Page (Cont'd) – SI (Metric) Units

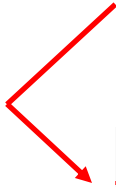
0.0	23.3997	79850.0	34020.0	2
61.38	12.2431	14.69	0	
61.39	21.8264	28965.0	15212.0	2
64.4	5.0688	14.69	0	
125.8	14.748	13409.0	4776.0	2
74.5	6.268	0.0	0	
510.0	14.748	3004.0	1305.0	2
67.3	2.3371	0.0	0	
0.0	23.3997	79850.0	34020.0	2



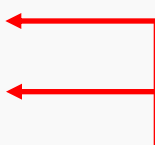
10th Input Page (Cont'd) – SI (Metric) Units

If Rocket Engine Model Type = 0
(No Rocket Engine, Glider
or Coasting Rocket),

Then These Lines are Empty.



0.0	23.3997	79850.0	34020.0	0
61.38	12.2431	14.69	0	0
61.39	21.8264	28965.0	15212.0	0
64.4	5.0688	14.69	0	0
125.8	14.748	13409.0	4776.0	0
74.5	6.268	0.0	0	0
510.0	14.748	3004.0	1305.0	0
67.3	2.3371	0.0	0	0



These are the

Stage	Stage	Stage	Stage	Rocket
Start Time	Aerodynamic	Initial	Burnout	Engine
(sec)	Reference	Weight	Weight	Model
	Area	(kg)	(kg)	Type
	(m^2)			

Inputs

10th Input Page (Cont'd) – SI (Metric) Units – Rocket Engine Model Type = 1

For Stage 1: This will dynamically update. (Stage 1, Stage 2, etc.)

Stage Data Make these Bold

Stage Start Time (sec)	Stage Aerodynamic Reference Area (m^2)	Stage Initial Weight (kg)	Stage Burnout Weight (kg)	Rocket Engine Model Type
				0

If Rocket Engine Model Type = 1
(Rocket Engine Model using
Chamber Pressure),
Then These Inputs.

- 0 - No Rocket Engine, Glider or Coasting Rocket
1 - Rocket Engine Model using Chamber Pressure
2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Throat Area (m^2)	Nozzle Expansion Ratio (n.d.)	Nozzle Divergence Half Angle (deg)	Burn Time (sec)	Reference Thrust (N)	Reference Specific Impulse (sec)	Reference Chamber Pressure (MPa)	Reference Atmospheric Pressure (MPa)
	0.0000	0.0000					0.0000

Ratio of Specific Heats (n.d.)	Thrust Coefficient Ratio (n.d.)	Nozzle Type 1 – Conical 2 – Bell	Negative Thrust 0 – Positive Thrust 1 – Negative Thrust
0.0000	0.0000	1	0

For Blank Boxes, Start with Box Blank. Allow User
to dynamically use .0000 to .0 Limit is 9000000.0

For Boxes with 0.0000, 4 digits, .0000

Have display start with Zero (Positive Thrust).
User has to change it (if they want to).

Integer

For Stage 1:

10th Input Page (Cont'd) – SI (Metric) Units – Rocket Engine Model Type = 1 Rocket Engine Model using Chamber Pressure

Stage Data

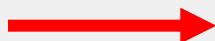
Stage Start Time (sec)	Stage Aerodynamic Reference Area (m ²)	Stage Initial Weight (kg)	Stage Burnout Weight (kg)	Rocket Engine Model Type
				0

0 - No Rocket Engine, Glider or Coasting Rocket
1 - Rocket Engine Model using Chamber Pressure
2 - Thrust Time History Rocket Engine Model

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Throat Area (m ²)	Nozzle Expansion Ratio (n.d.)	Nozzle Divergence Half Angle (deg)	Burn Time (sec)	Reference Thrust (N)	Reference Specific Impulse (sec)	Reference Chamber Pressure (MPa)	Reference Atmospheric Pressure (MPa)
	0.0000	0.0000					0.0000

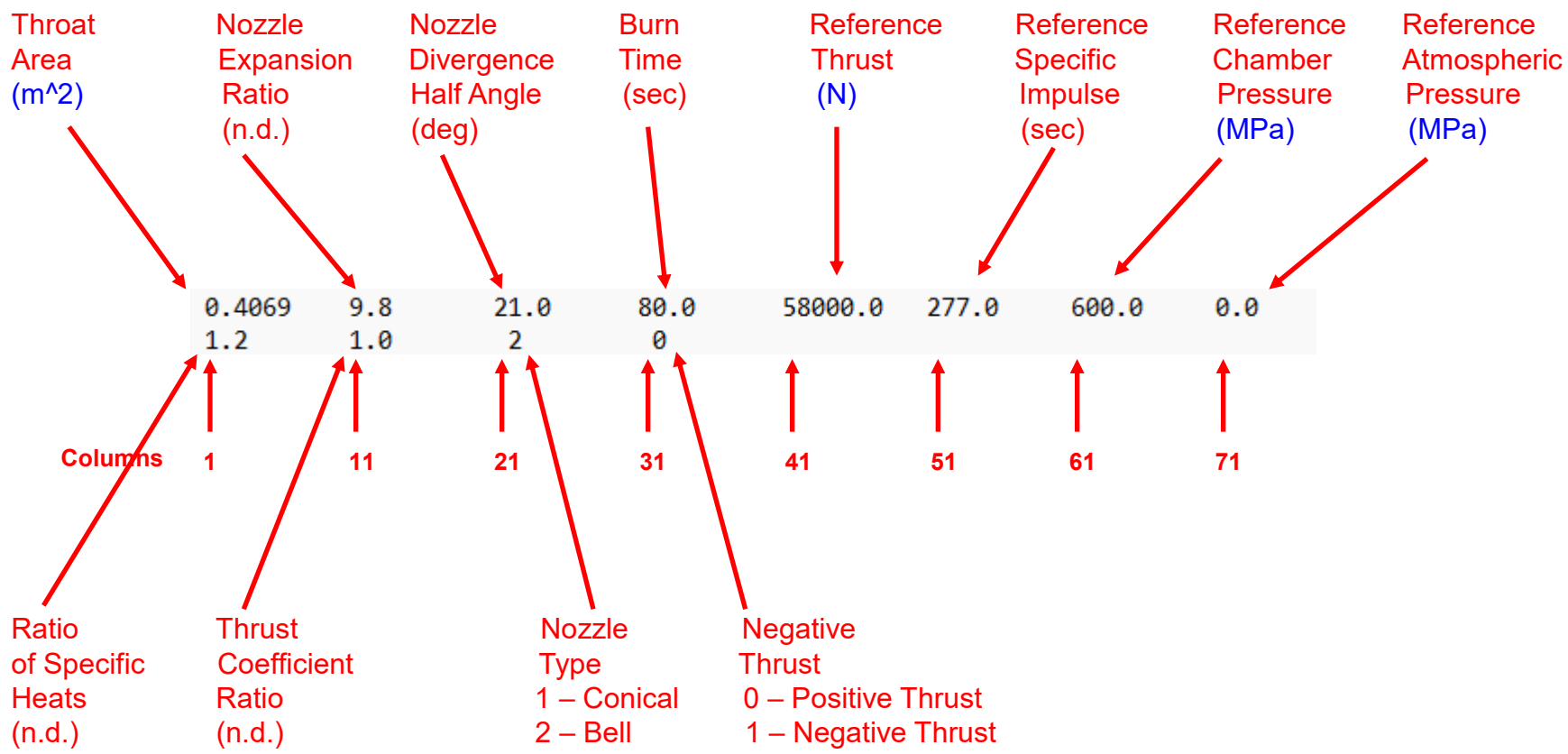
Ratio of Specific Heats (n.d.)	Thrust Coefficient Ratio (n.d.)	Nozzle Type 1 – Conical 2 – Bell	Negative Thrust 0 – Positive Thrust 1 – Negative Thrust
0.0000	0.0000	1	0



Save

Cancel

10th Input Page (Cont'd) – SI (Metric) Units – Rocket Engine Model Type = 1
Rocket Engine Model using Chamber Pressure



This was Rocket Engine Model = 1 Rocket Engine Model using Chamber Pressure

If Rocket Engine Model Type = 0 No Rocket Engine, Glider or Coasting Rocket),

Then These Input Lines Would Not Be Present.

10th Input Page (Cont'd) – SI (Metric) Units – Rocket Engine Model Type = 2

For Stage 1: This will dynamically update. (Stage 1, Stage 2, etc.)

Stage Data Make these Bold

Stage Start Time (sec)	Stage Aerodynamic Reference Area (m ²)	Stage Initial Weight (kg)	Stage Burnout Weight (kg)	Rocket Engine Model Type
				0

If Rocket Engine Model Type = 2
(Thrust Time History Rocket Engine
Model),

- 0 - No Rocket Engine, Glider or Coasting Rocket
- 1 - Rocket Engine Model using Chamber Pressure
- 2 - Thrust Time History Rocket Engine Model

Then These Inputs.

Rocket Engine can be Liquid Rocket Engine or Solid Rocket Motor

Burn Time (sec)	Nozzle Exit Area (m ²)	Reference Atmospheric Pressure (MPa)	Negative Thrust 0 – Positive Thrust 1 – Negative Thrust
		0.0000	0

For Blank Boxes, Start with Box Blank. Allow User
to dynamically use .0000 to .0 Limit is 9000000.0

For Boxes with 0.0000, 4 digits, .0000

Integer

Have display start with Zero (Positive Thrust).
User has to change it (if they want to).

Save

Cancel

10th Input Page (Cont'd) – SI (Metric) Units – Rocket Engine Model Type = 2
Thrust Time History Rocket Engine Model

These are the

Stage Start Time (sec)	Stage Aerodynamic Reference Area (m ²)	Stage Initial Weight (kg)	Stage Burnout Weight (kg)	Rocket Engine Model Type
------------------------------	--	------------------------------------	------------------------------------	-----------------------------------

Inputs							
2							
0.0	0.0						
0.0	0.0						
0.0	0.0						
0.0	0.0	0.0					
1	180.0	1	5.9				
385.0	0.0	34.5764	-120.6324	90.0	180.0	0.0	0.0
0.01	578.0	1.0					
0.0	23.3997	79850.0	34020.0	2			
61.38	12.2431	14.69	0				

Columns

1

11

21

31

32

0

1

Burn Time (sec)

Nozzle Exit Area (m²)

Reference Atmospheric Pressure (MPa)

Negative Thrust
0 – Positive Thrust
1 – Negative Thrust

11th Input Page – SI (Metric) Units – Rocket Engine Model Type = 1
Chamber Pressure Time History

Chamber Pressure Time History

Make these Bold

Number of Chamber Pressure Time History Points

Minimum of 2, Maximum of 40, is an Integer

Time (sec)

First Time Must Be Zero

Chamber Pressure (MPa)

Will dynamically update based on number of Chamber Pressure Time History Points. 8 Boxes per Row until Final Row, Final Row will have Number to Match Number of Inputs.

Blank Boxes, Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 No Negative Numbers.

Save

Cancel

11th Input Page (Cont'd) – SI (Metric) Units – Rocket Engine Model Type = 1
Chamber Pressure Time History

First Time Must Be Zero

Number of Chamber Pressure Time History Points
Integer
If > 9, then two digits (38)
If < 10, then _3 _8 _9 etc., put one space in front of Integer

Time (sec)

Chamber Pressure (MPa)

3		
0.0	79.9	80.0
600.0	600.0	0.0

Columns 1 11 21

11th Input Page (Cont'd) – SI (Metric) Units – Rocket Engine Model Type = 2
Thrust Time History (Thrust Curve)

Thrust Time History

Number of Thrust Time History Points

38

Minimum of 2, Maximum of 40, is an Integer

Time (sec)

Thrust (N)

First Time Must Be Zero

Will dynamically update based on number of Chamber Pressure Time History Points. 8 Boxes per Row until Final Row, Final Row will have Number to Match Number of Inputs.

Blank Boxes, Start with Box Blank. Allow User to dynamically use .0000 to .0 Limit is 9000000.0 No Negative Numbers.

Save

Cancel

11th Input Page (Cont'd) – SI (Metric) Units – Rocket Engine Model Type = 2
Thrust Time History (Thrust Curve)

First Time Must Be Zero

Number of Thrust Time History Points

Integer

If > 9, then two digits (38)

If < 10, then _8 _9 etc., put one space in front of Integer

	38							
Time (sec)	0.0	0.71	5.40	15.94	31.76	47.59	52.86	57.28
	61.38	61.39	66.99	72.07	77.14	87.30	97.46	112.69
	124.40	125.79	125.8	140.8	155.8	160.8	165.8	180.8
	191.59	195.8	200.3	510.0	510.01	513.1	525.0	533.0
	540.0	567.0	568.6	574.8	577.3	578.0		
Thrust (N)	102186.7	199786.7	198788.4	203779.8	200170.6	201860.0	200093.9	45822.8
	0.0	42654.1	51364.6	55211.3	56928.0	56355.7	50379.1	44211.8
	30446.6	0.0	22250.0	37105.0	42949.0	43846.0	42949.0	37632.0
	28590.0	2308.0	0.0	0.0	5213.0	6133.3	7783.0	7783.0
	8255.5	8255.5	6000.0	4000.0	0.0	0.0		
Columns	1	11	21	31	41	51	61	71